

Final Individual Project Report

MGMT479 Data Visualization
Section 1:30

Evaluating Competitive Advantage and Performance Dominance in Formula 1

Team Members:

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Primary Dataset:

[Formula 1 Race Data \(1950–2017\)](#), Kaggle

Business Background

Behind every Formula 1 victory lies far more than speed alone. In one of the world's most complex and data-rich competitive contexts, Formula 1 success is influenced not only by driver skill but also by constructor engineering, race strategy, pit execution, circuit fit, and regulatory adaptation. Teams have produced decades' worth of race and standings data since 1950, but a large portion of that data is still dispersed throughout seasons, drivers, and tracks rather than being combined into insightful strategic analysis. Historical performance data is only useful for firms who compete at the top level of racing when it can be converted into patterns that guide future choices. **What makes Formula 1 especially compelling from a business perspective is that it mirrors long-term competitive strategy in any high-stakes industry.** Constructors must constantly decide where to invest, whom to recruit, how to respond to rivals, and when to adapt their operating model to maintain an edge. In that sense, Formula 1 is not just a sport; it is a global performance business driven by innovation, timing, and precision. This project uses the Formula 1 Race Data (1950–2017) dataset to turn decades of competition into executive-level visual intelligence, revealing how dominance is built, sustained, challenged, and sometimes lost over time.

Problems for the Target Audience

The target audience for this project is Formula 1 constructor leadership and performance strategy teams, especially decision-makers at organizations such as Mercedes, Ferrari, Red Bull, and McLaren. These stakeholders make high-impact decisions about engineering investment, driver selection, race preparation, and long-term planning. Although they have access to race results and standings, the raw data does not immediately show which teams built sustainable dominance, where performance advantages were strongest, or how success shifted across time. Several questions remain difficult to answer without structured visualization. Race outcomes alone do not clearly reveal how competitive eras changed or which constructors maintained an edge across multiple seasons. It can also be difficult to separate the influence of driver talent from the strength of the team behind that driver. Circuit-specific performance patterns, qualifying outcomes, pit strategy, and finishing consistency may all shape results, yet those relationships are easy to miss in spreadsheets or isolated tables. A visual analysis is therefore necessary to organize the data into a form that supports evidence-based strategy rather than broad historical impression.

Analysis: Project Objectives

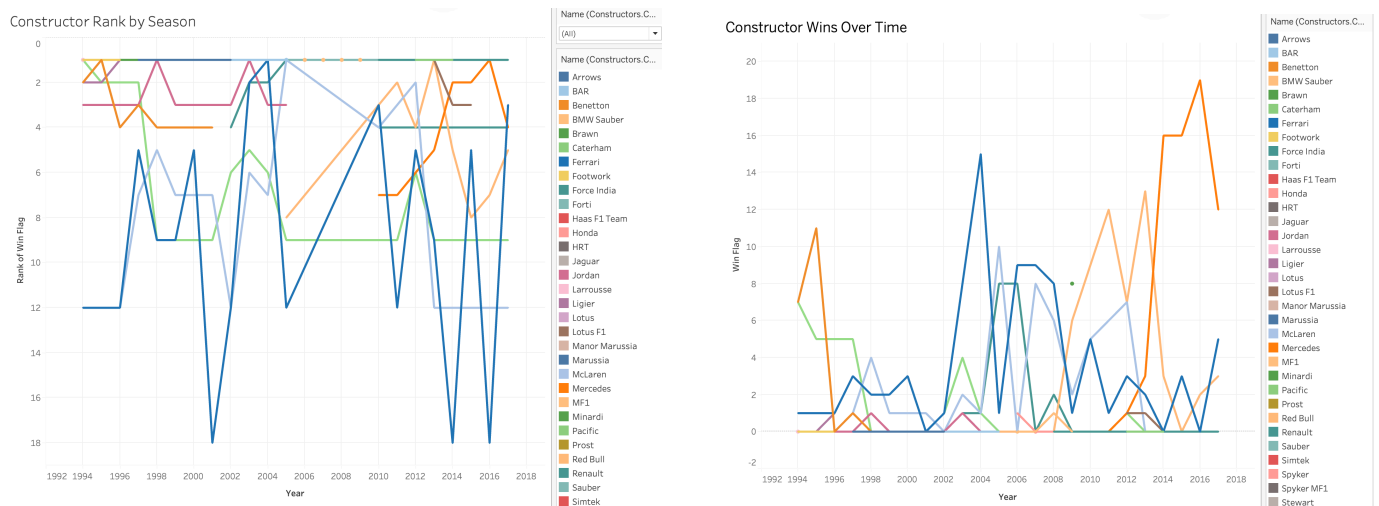
The objective of this project is to evaluate how competitive advantage in Formula 1 is created, sustained, and shifted across teams, drivers, circuits, and eras. The analysis focuses on the deeper structure of performance rather than a simple summary of race wins or championships. It examines which constructors led different periods of the sport, how much of that success can be associated with drivers versus teams, whether specific circuits favored certain constructors, and how race-related measures such as qualifying position and finishing outcomes contribute to

broader performance patterns. A second objective is to translate a large historical dataset into a form that is useful for decision-makers. The project is designed to present Formula 1 history as a strategic story supported by multiple chart types and a Tableau dashboard. By doing so, it helps the audience interpret dominance cycles, identify areas of strength and weakness, and understand which forms of success appear durable over time.

Data Analysis / Visualization Discussion

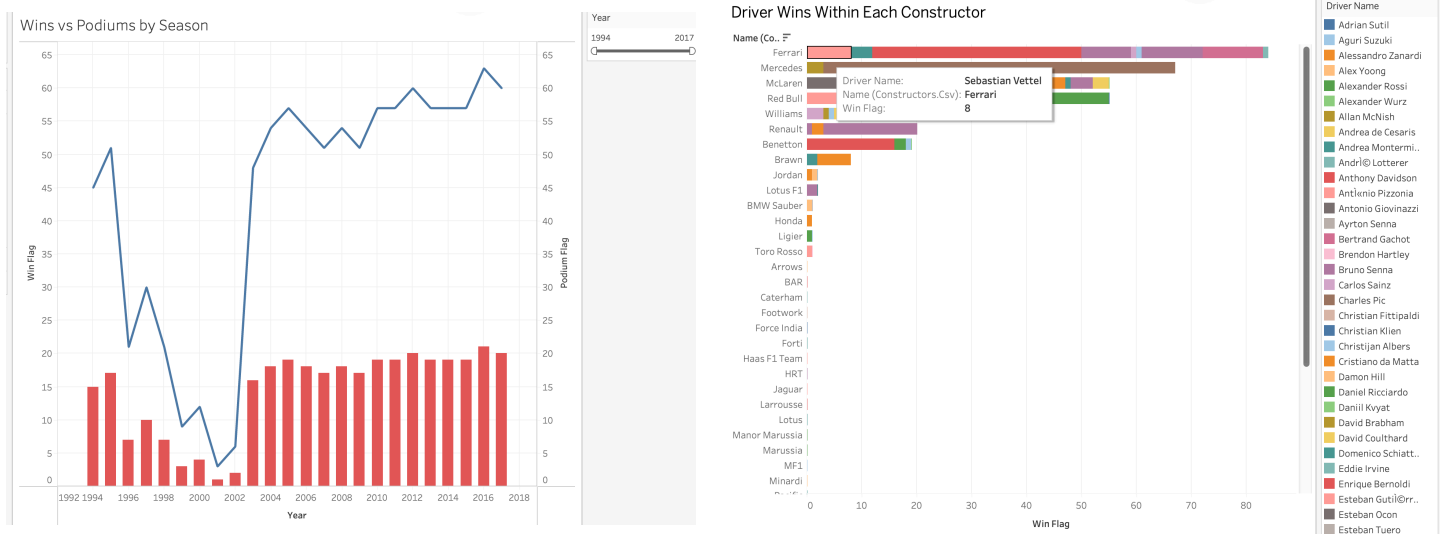
This analysis integrates multiple visualizations to examine how competitive advantage in Formula 1 is built, sustained, and translated into race outcomes. Each chart contributes a distinct layer of insight, collectively moving from long-term dominance patterns to race-level execution and global context.

The **Constructor Wins Over Time** chart establishes the foundation of the analysis by quantifying dominance cycles across seasons. By tracking total wins per constructor annually, the visualization highlights clear eras of control, where specific teams sustain elevated win counts relative to competitors. These peaks indicate periods of superior engineering, strategy, and resource allocation. Complementing this, the **Constructor Rank by Season** chart introduces positional consistency as a performance metric. While win counts capture peak success, ranking trends reveal stability; teams that maintain top rankings across consecutive seasons demonstrate sustained competitive advantage rather than isolated success.

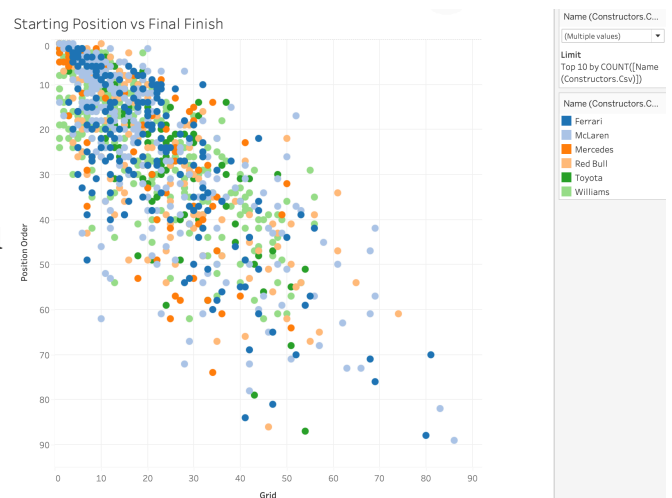


Building on this, the **Driver Wins Within Each Constructor** chart decomposes total wins into driver-level contributions. By aggregating wins by driver within each team, the visualization distinguishes whether success is concentrated among a few elite drivers or distributed across multiple contributors. This allows for a clearer interpretation of whether performance is primarily driven by individual talent or by the strength of the constructor's system. The **Wins vs. Podiums by Season** chart extends this analysis by comparing total victories to total podium finishes. The gap between these two metrics serves as an indicator of performance efficiency, teams with high podium counts but fewer wins demonstrate consistency without

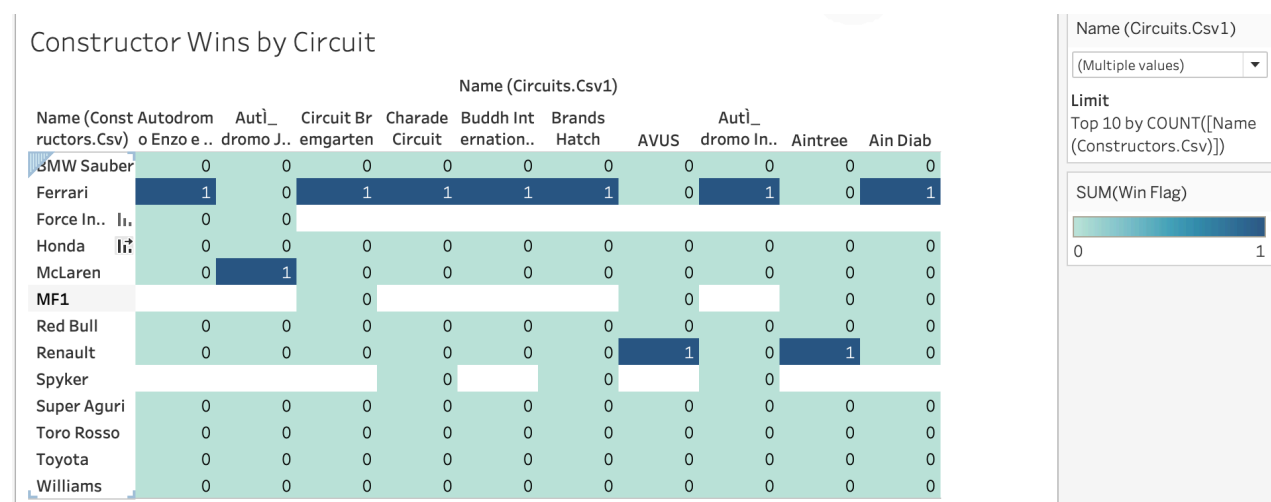
dominance, whereas teams with a higher win-to-podium ratio exhibit stronger race conversion capability.



At the race execution level, the **Starting Position vs. Final Finish** scatter plot evaluates the relationship between qualifying performance (grid position) and race outcomes (final position). The observed negative correlation indicates that lower grid positions (closer to pole) are generally associated with better finishing results, reinforcing the importance of qualifying strategy. However, the dispersion of points also reveals variability, suggesting that race-day factors such as pit strategy, overtaking ability, and reliability significantly influence final outcomes beyond starting position alone.

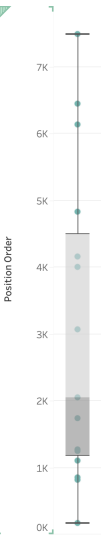


To assess circuit-specific performance, the **Constructor Wins by Circuit** heatmap visualizes win frequency across different tracks. By mapping constructors against circuits, the chart highlights areas of specialization, where certain teams consistently outperform others in specific environments. These patterns suggest that track characteristics, such as layout, weather, and



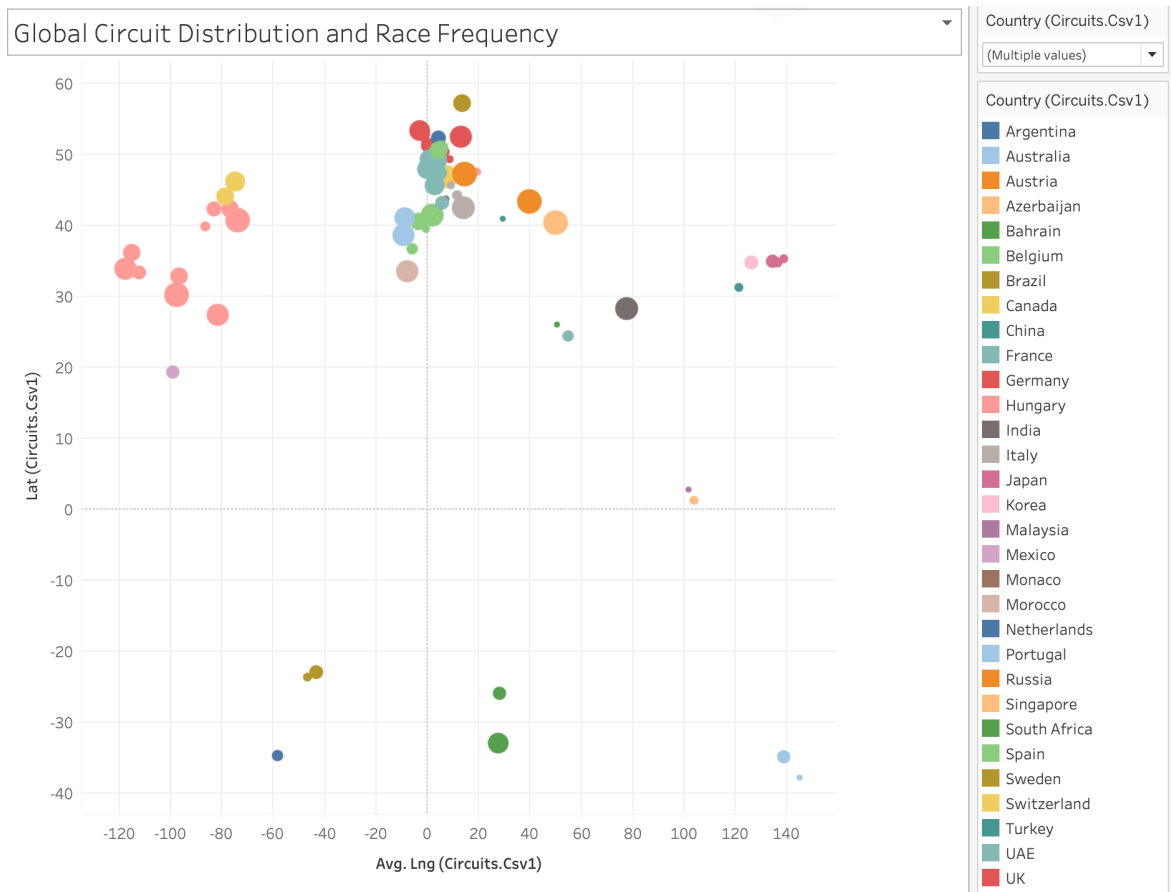
technical demands, interact with team design and strategy, creating localized competitive advantages.

Distribution of Finishing Positions by Constructor



The analysis then shifts to a **boxplot of finishing-position distributions by constructor**. At this point in the story, the focus moves from moments of strong performance to the consistency of those results across races. A constructor with a strong median finish and a narrower range of outcomes may be interpreted as steady and dependable. A wider spread suggests greater volatility, even when wins are present. This chart helps show that sustained success in Formula 1 depends on more than occasional victories. Consistency across an entire season often shapes the difference between a competitive team and a championship-caliber one.

Finally, the **Global Circuit Distribution and Race Frequency** visualization provides geographic context by mapping race locations and frequency across countries. This chart illustrates the expansion of Formula 1 into diverse global markets and highlights regions with higher race concentration. The distribution not only reflects the sport’s commercial growth but also introduces environmental and logistical variability that can influence team performance.



Concluding Remarks

This analysis resolves the initial challenge of fragmented race data by consolidating multiple performance dimensions into a cohesive, visualization-driven narrative. By linking constructor dominance, driver contribution, race execution, and geographic context, the dashboard provides a comprehensive view of how competitive advantage is formed and sustained in Formula 1.

From a strategic perspective, the findings highlight that long-term success is not driven by a single factor but by the alignment of consistent rankings, efficient race conversion, and adaptability across circuits. Constructors that combine high win counts with stable top rankings demonstrate the strongest competitive positioning, while the relationship between podium frequency and wins underscores the importance of converting opportunities into victories. Additionally, the variation observed in race outcomes relative to starting positions reinforces the role of in-race decision-making and execution as differentiators beyond qualifying performance.

For the target audience of constructor executives and performance strategists, this analysis delivers actionable value by identifying patterns that inform decision-making. Teams can use these insights to evaluate driver effectiveness within their systems, optimize race strategies to improve win conversion rates, and allocate engineering resources toward circuits where competitive gaps exist. Furthermore, the geographic analysis provides context for strategic planning in a globally expanding sport, where regional performance and adaptability are increasingly relevant.

Ultimately, the dashboard transforms historical data into decision-oriented intelligence, enabling stakeholders to move beyond anecdotal interpretation and toward evidence-based strategy. This structured approach enhances the ability to forecast performance trends, refine competitive positioning, and sustain long-term success in Formula 1.

